

machines or robots with entirely new natural user interfaces. Finally, these and many more could be merged to a full-featured Li-Fi cloud providing wireless services for other future applications as well.

What Li-Fi stands for

Li-Fi comprises a wide range of frequencies and wavelengths, from the infrared through visible and down to the ultraviolet spectrum. It includes sub-gigabit and gigabit-class communication speeds for short, medium and long ranges, and unidirectional and bidirectional data transfer using line-of-sight or diffuse links, reflections and much more. It is not limited to LED or laser technologies or to a particular receiving technique. Li-Fi is a framework for all of these providing new capabilities to current and future services, applications and end users.

Usage models

Within a local Li-Fi cloud several data-based services are supported through a heterogeneous communication system. In an initial approach, the Li-Fi Consortium defined different types of technologies to provide secure, reliable and ultra-high-speed wireless communication interfaces. These technologies included giga-speed technologies, optical mobility technologies, and navigation, precision location and gesture recognition technologies.

For giga-speed technologies, the Li-Fi Consortium defined GigaDock, GigaBeam, GigaShower, GigaSpot and GigaMIMO models (Fig. 2) to address different user scenarios for wireless indoor and indoor-like data transfers. While GigaDock is a wireless docking solution including wireless charging for smartphones,

tablets or notebooks, with speeds up to 10 Gbps, the GigaBeam model is a point-to-point data link for kiosk applications or portable-to-portable data exchanges. Thus a two-hour full HDTV movie (5 GB) can be transferred from one device to another within four seconds.

GigaShower, GigaSpot and GigaMIMO are the other models for in-house communication. There a transmitter or receiver is mounted into the ceiling connected to, for example, a media server. On the other side are portable or fixed devices on a desk in an office, in an operating room, in a production hall or at an airport. GigaShower provides unidirectional data services via several channels to multiple users with gigabit-class communication speed over several metres. This is like watching TV channels or listening to different radio stations where no uplink channel is needed. In case GigaShower is used to sell books, music or movies, the connected media server can be accessed via Wi-Fi to process payment via a mobile device. GigaSpot and GigaMIMO are optical wireless single- and multi-channel HotSpot solutions offering bidirectional gigabit-class communication in a room, hall or shopping mall for example.

How Li-Fi works

Imagine yourself walking into a mall where GPS signals are unavailable but the mall is equipped with ceiling bulbs that create their own 'constellation' of navigation beacons. As the camera of your cellphone automatically receives these signals, it switches your navigation software to use this information to guide you to the ATM machine you're looking for.

You conclude your ATM transaction and notice the GigaSpot sign for instant digital movie downloads. You pick out that new Tom Cruise movie using your phone's payment facility, and then download within a few seconds the high-definition movie into the GigaLink flash drive plugged into the USB port of your smartphone.

As you walk away, your phone notifies you that the leather jacket Tom featured in the movie is on sale nearby. You walk over towards the show window and your image comes up on the screen, wearing that coveted jacket. You turn and pose while the image matches your orientation and body gestures for a 'digital fitting.' When you walk into the store, the clerk hands you the actual jacket in exactly your size.

On the verge of a breakthrough

First applications of Li-Fi have been put to use already, for example, in hospitals where RF signals are a threat due to interference problems with medical equipment such as blood pumps and other life supporting instruments. Axiomtek Europe presented such a product at the *Embedded World* exhibition in Nürnberg, Germany. The prototype of a mobile phone with an incorporated VLC system was presented by Casio at the *Consumer Electronics Show* in Las Vegas in January this year. In the coming years, we will see more Li-Fi products entering the market, both in the industrial as well as consumer markets. ●

The authors are co-founders of the Li-Fi Consortium (www.lificonsortium.org)—a non-profit organisation focusing on optical wireless technologies and fostering the Li-Fi cloud in general

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